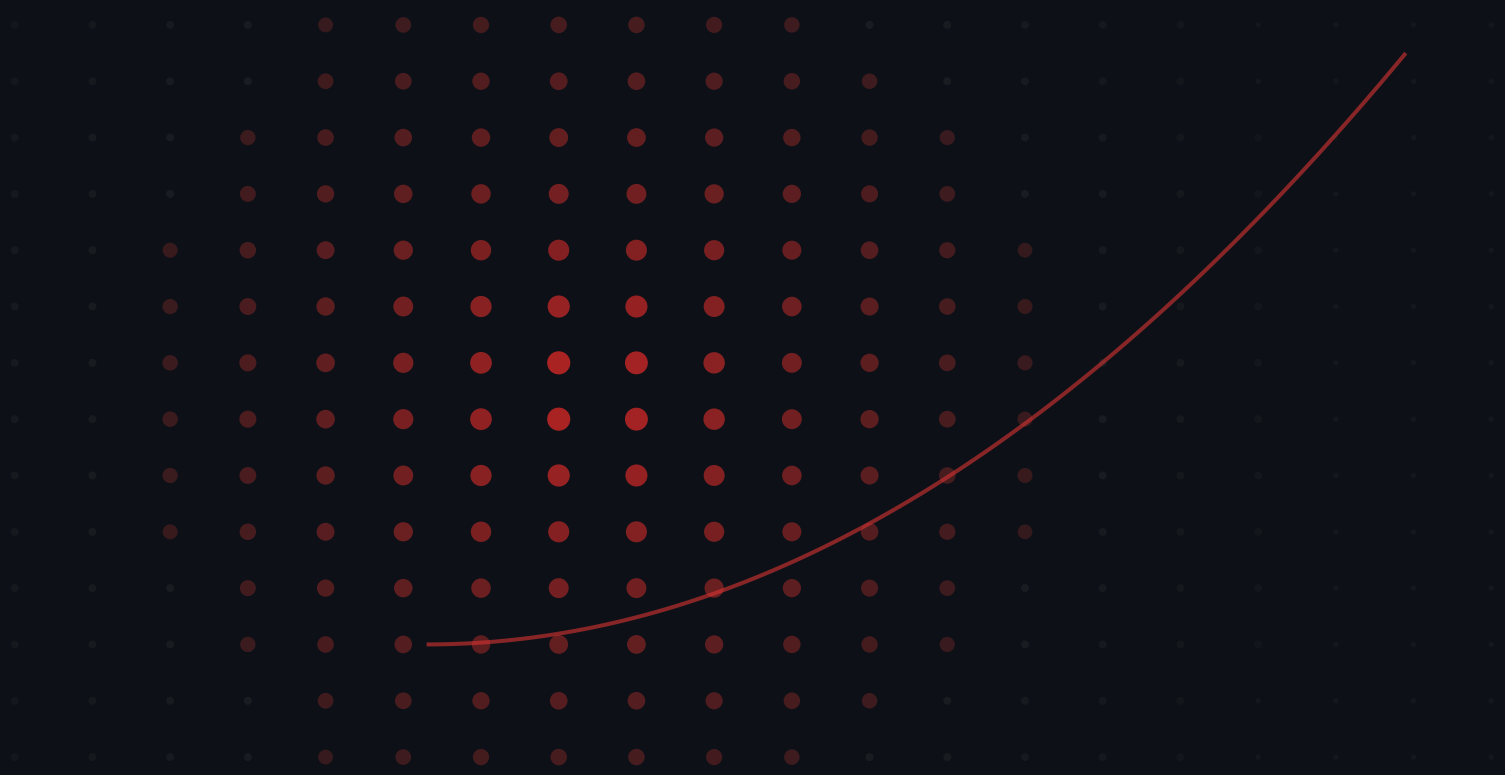
GREYNOISE

The Invisible Army: Residential Proxy Abuse in Internet-Scale Attack Traffic

Why IP Reputation Fails Against the Rotation Economy

90 days | 4 billion sessions



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Why This Matters

GreyNoise observed 4 billion malicious sessions targeting the edge over 90 days. The data reveals a pattern that challenges a core assumption of network defense: that you can tell attackers from legitimate users by where the traffic comes from.

A residential proxy is a compromised home internet connection used as a disguise. Attackers route malicious traffic through ordinary home broadband, mobile data, and small-business connections — the same IP address ranges used by employees, customers, and partners. To a firewall or reputation feed, the traffic looks indistinguishable from a normal user.

In January 2026, Google Threat Intelligence Group disrupted IPIDEA — one of the largest residential proxy networks, with 9 to 11 million daily active proxies used by over 550 distinct threat groups, including state-sponsored actors from China, DPRK, Iran, and Russia. The FBI dismantled 911 S5 in May 2024 — 19 million IPs across 190 countries. Mandiant M-Trends 2025 documented state actors routing operations through residential infrastructure.

The problem is not theoretical. Every enterprise firewall processes traffic from residential IP space. Every reputation feed fails to flag IPs that rotate before they can be cataloged. This report quantifies the residential proxy threat at internet scale and identifies what defenders can do about it. This paper presents a sample of the consumer-network activity visible in GreyNoise data.

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Key Takeaways

1. 39% of unique IPs targeting the edge come from home internet connections. That is nearly double their 22% share of sessions — each residential IP averages fewer than 3 sessions before disappearing, and the median is just 1.

2. 78% are invisible to reputation feeds. Nearly four out of five residential IPs are observed at most twice across the entire global sensor network before rotating. They vanish before any reputation system can flag them.

3. Compromised home PCs follow the human sleep cycle. Traffic from IPs geolocating to India drops 34% at night — the infected machines are physically powered off. Server traffic varies less than 3%.

4. Residential IPs scan — they almost never exploit. Only 0.1% of residential sessions carry exploitation payloads, versus 1.0% from hosting infrastructure. Residential proxies map the terrain; the exploitation payloads come later from datacenter infrastructure, suggesting a division of labor where residential nodes perform reconnaissance and hosting-based operators act on the results.

5. Four separate threats hide behind one label. Worm propagation, IoT botnets, commercial proxy fleets, and VPN reconnaissance. At minimum two of these — worm propagation and IoT botnets — show zero IP overlap, confirming they are completely separate populations.

6. When one proxy network dies, another takes its place. After IPIDEA lost 40% of its nodes, operators backfilled with datacenter servers within weeks. According to IPinfo research, 46% of proxy IPs span multiple providers simultaneously.

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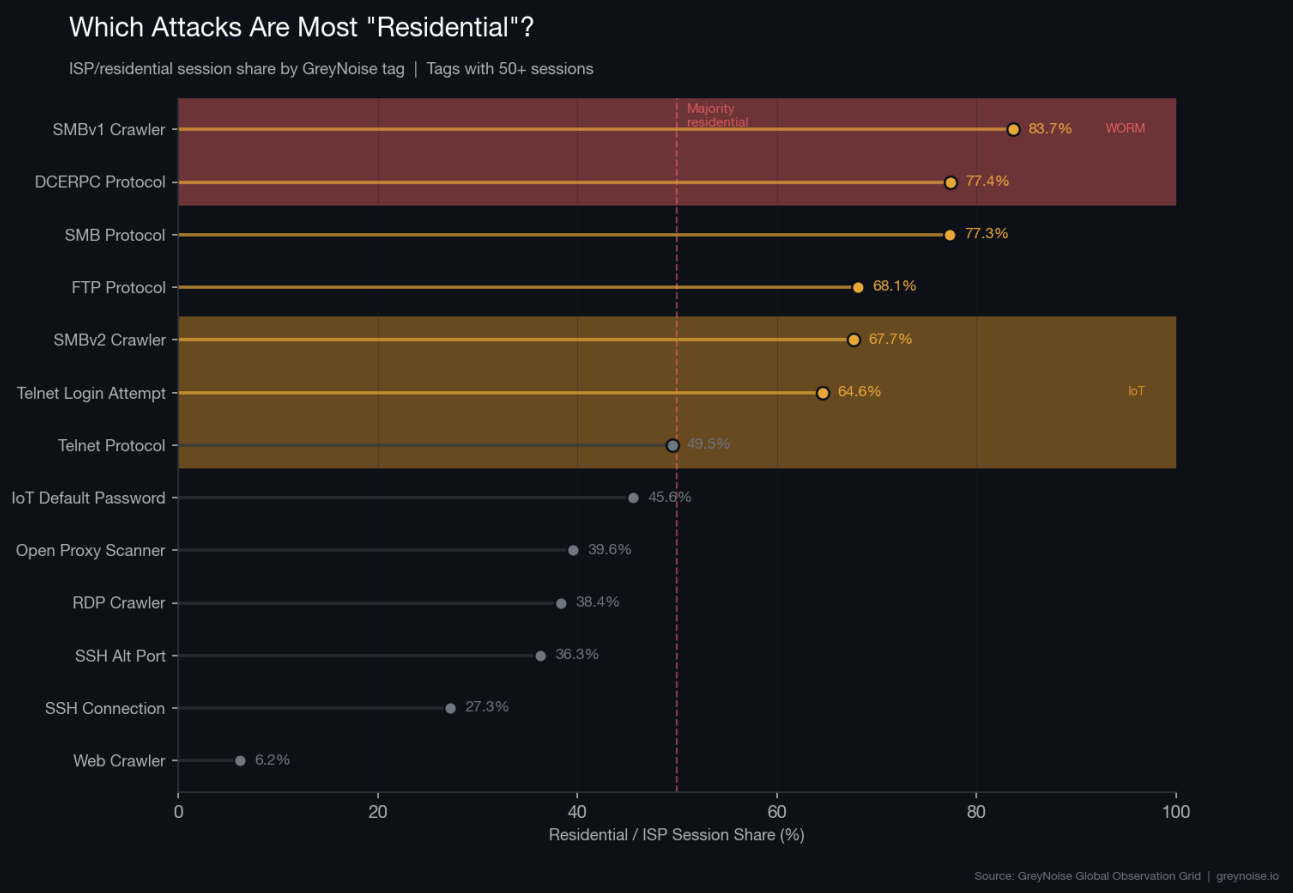
The Landscape

Nearly 4 in 10 unique IPs targeting the edge originate from residential address space — and they behave nothing like datacenter attackers.



Residential IPs fire a few sessions and vanish. 2.7 sessions each on average, versus 5.9 for datacenter IPs. They account for 39% of unique IPs observed but only 22% of total sessions. They are everywhere, briefly.

IPinfo classifies IPs into types including "hosting" (cloud and datacenter infrastructure) and "isp" (consumer broadband, mobile, and small-business connections). We use "residential" as shorthand for ISP-classified address space throughout this report.

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IP Type	Sessions	Session %	Unique IPs	IP %	Sess/IP
Hosting	22,527	75%	3,791	59%	5.9
ISP (Residential)	6,734	22%	2,487	39%	2.7
Business	644	2%	143	2%	4.5
Education	75	<1%	38	1%	2

Table 1: IP type distribution across 30,000-session sample (10K/month, Dec 2025 – Feb 2026). Sessions-per-IP ratios are upper bounds on per-device activity in countries with heavy carrier-grade NAT (CGNAT) deployment, where multiple devices share a single public IP.

Session share is estimated from a 30,000-session sample and carries meaningful uncertainty (the 22% estimate ranges from 15% to 30% depending on sampling method). The IP share of 39% is more stable across sampling methods and is the more reliable figure.

Not all IPs classified as residential are genuinely residential. Our Censys ground-truth validation of 50 IPs found 42% were genuinely residential endpoints; another 20% were compromised CPE devices (routers, gateways) at residential locations; the remaining 38% were misclassified servers or scanners. The behavioral findings in this report — circadian patterns, zero protocol overlap, rotation velocity — hold regardless of the exact residential proportion, but absolute percentages should be interpreted as upper bounds on true residential share.

Legacy protocols — SMB, DCERPC, Telnet — are overwhelmingly residential. SSH, TLS, and RDP scanning runs primarily from hosting infrastructure. Web Crawler shows the lowest residential share (6.2%), but this figure reflects a methodological blind spot: GreyNoise sensors are honeypots with no real content to scrape, so web scraping — one of the largest commercial use cases for residential proxies, including LLM training data collection — is structurally invisible to this methodology. The true residential share of web scraping activity is almost certainly far higher than 6.2%.

> **WHAT THIS INDICATES**
>
> By the time a reputation feed flags a residential IP, the malicious behavior has already rotated to a new address. Detection must shift from "where is the traffic from?" to "what is the traffic doing?"

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The Rotation Economy

IP reputation is structurally broken against residential proxies. The math makes it impossible.

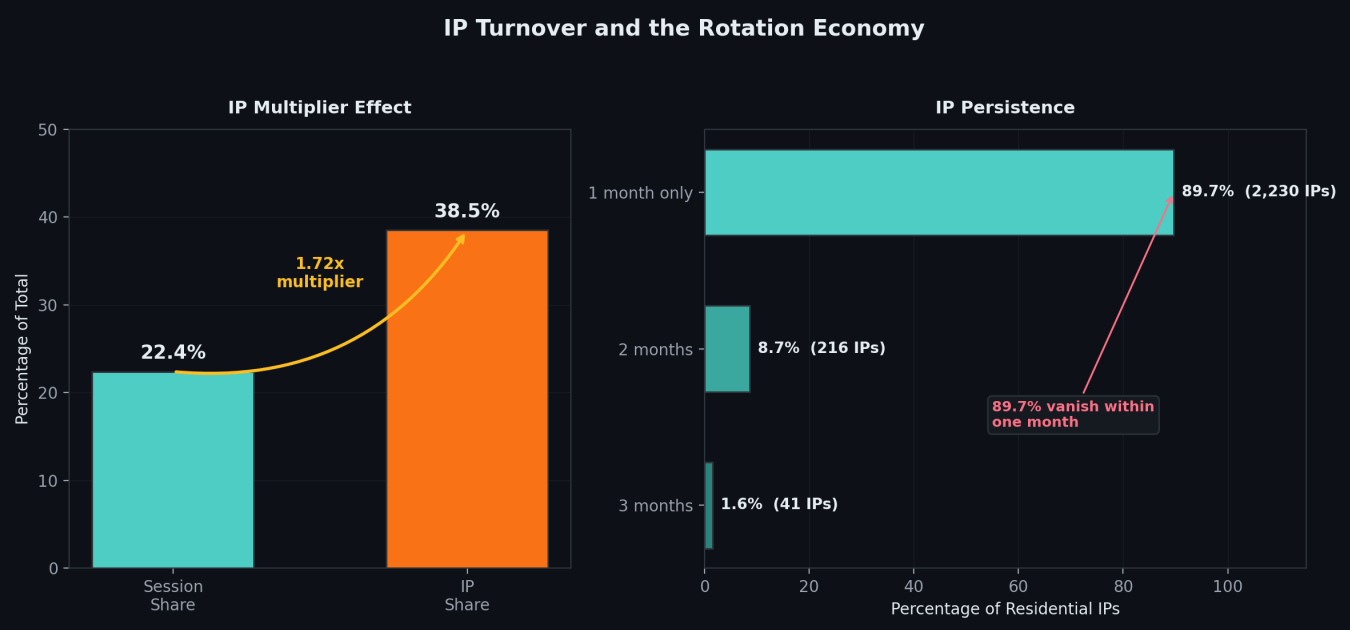

Figure 5: Residential IPs represent 22.4% of sessions but 38.5% of unique IPs — a 1.72x multiplier. 89.7% appear in only one month's sample, reflecting DHCP lease rotation, proxy cycling, or both.

78% of residential IPs appear in only 1–2 sessions and are never observed again. The typical one fires once and vanishes — the device gets a new IP from its ISP, or the proxy SDK routes traffic through a different exit node.

In our sample, 90% of residential IPs appeared in only one month — though the sparse sampling rate (10,000 sessions per month from millions) means even persistent IPs would often be captured in only one month's sample. The 41 IPs that persisted across all three months tell a different story: 100% SSH-focused, spanning 26 countries, with Linux TCP stacks — likely enrolled as long-term proxy endpoints.

Why This Is a Structural Problem

- Rapid rotation: 2.7 sessions per IP on average — rotated before reputation systems can flag them

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- Massive diversity: 683 distinct ISP organizations — no single network to block
- Sparse appearance: 78% appear in only 1–2 sessions across the global sensor network
- Legitimate dual-use: Real people use these same IPs — broad blocking causes collateral damage

> **WHAT THIS INDICATES**
> Static blocklists are structurally ineffective against residential proxies. Detection must move to behavioral analysis and device fingerprinting — signals that survive IP rotation.

The Sleep Cycle

Attack traffic from compromised home PCs follows the human sleep cycle — and GreyNoise data confirms it.

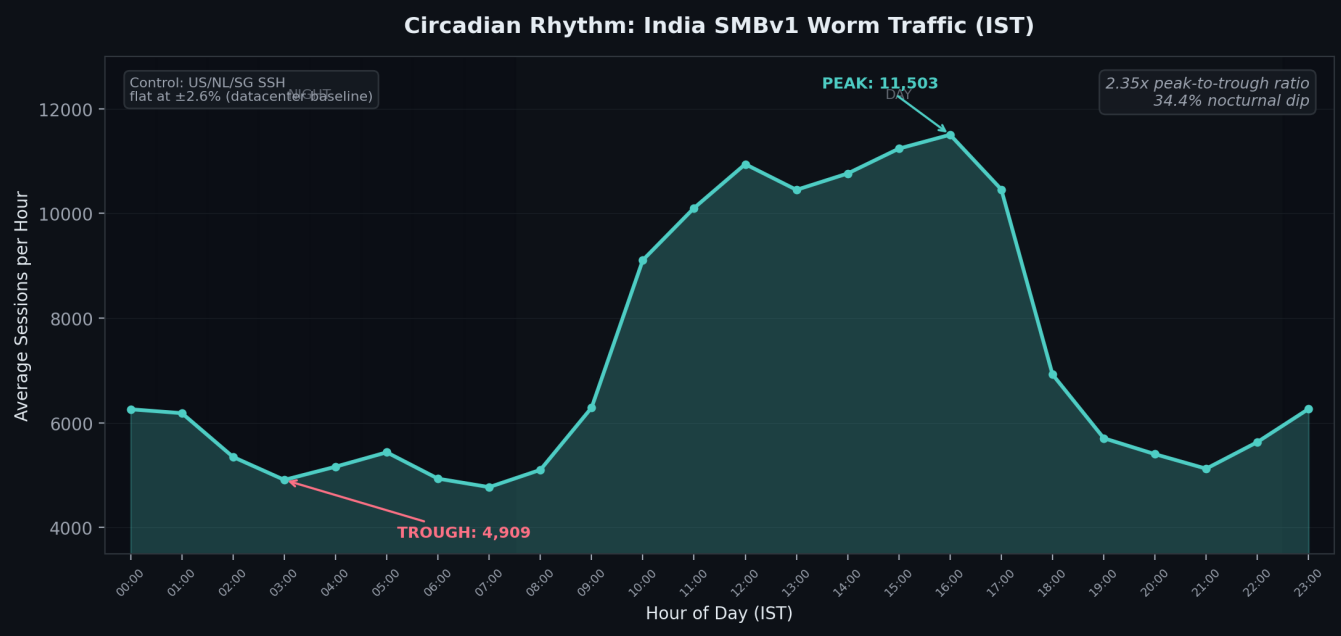

Figure 1: India SMBv1 worm traffic peaks at 4PM local time and drops 34% at night — compelling evidence these are home devices following human activity cycles.

Traffic from IPs geolocating to India drops 34% between daytime peak and overnight trough, measured across the 90-day observation window. The pattern is unmistakable: activity climbs when people wake up, peaks when home PCs are powered on, and drops sharply in the evening.

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This circadian analysis is specific to SMBv1 worm propagation, which runs on Windows PCs. The most likely explanation is that Indian home PCs are physically powered off overnight — a common practice driven by power costs and electricity reliability. India's large mobile device population (Jio, Airtel) would exhibit different idle patterns not captured by this analysis, as mobile devices may show proxy SDK pausing behavior or users disabling mobile data overnight rather than full power-off cycles. Alternative explanations include dynamic IP reassignment by ISPs and proxy SDK activity tied to device usage patterns. In the United States, where always-on computing is more common, the circadian pattern disappears entirely.

The control group validates the methodology. Server-based SSH traffic from US, Netherlands, and Singapore datacenters runs flat around the clock — less than 3% variation. Infrastructure that never sleeps shows no circadian signal. Compromised home PCs do.

> **WHAT THIS INDICATES**
> These are infected home PCs, not dedicated attack infrastructure. The malware can only scan when someone is home using their computer. The device owners are victims.

Where They Come From

683 ISP organizations across the globe contribute residential attack traffic — and no single network holds more than 8% of the total.

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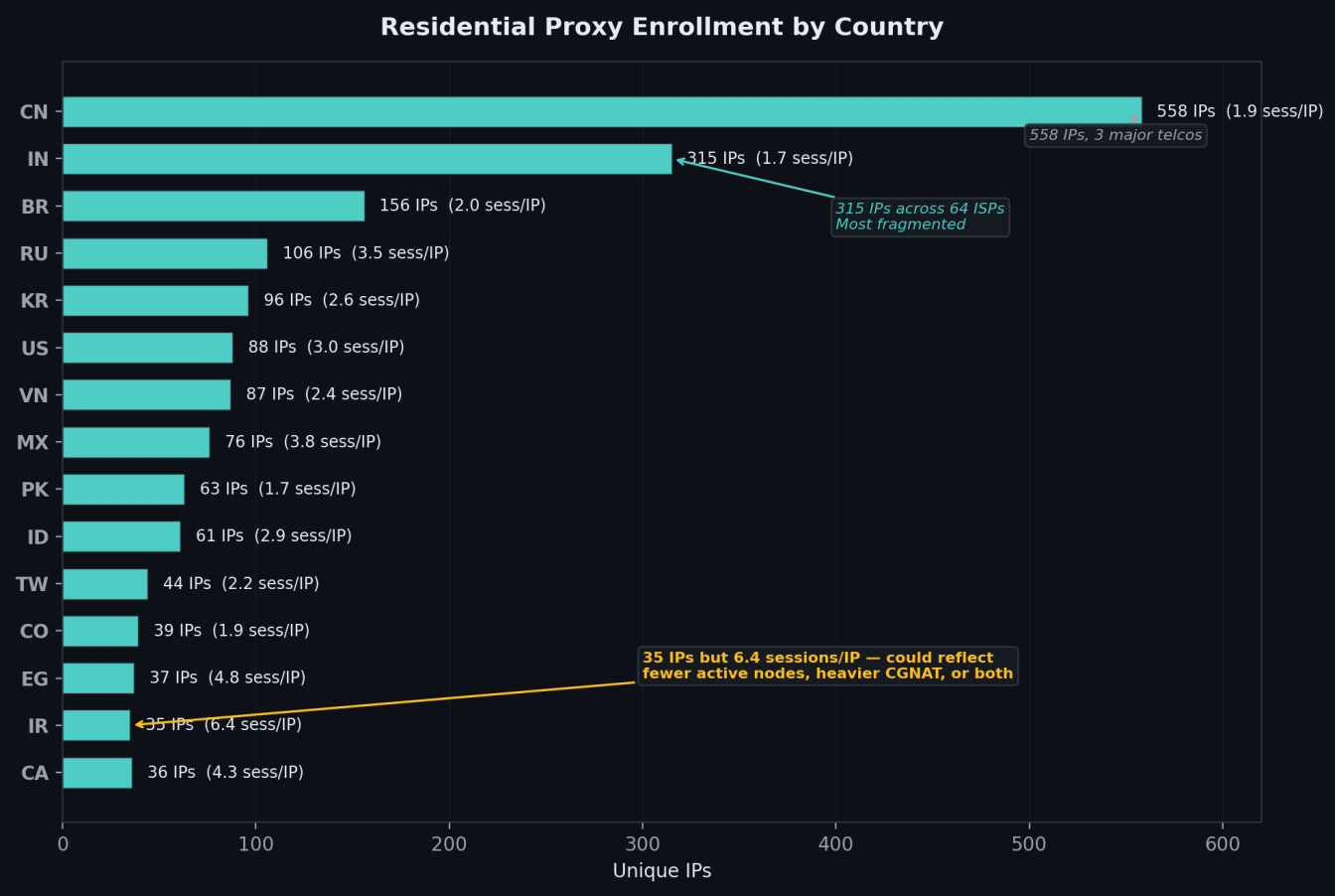

Figure 7: China dominates residential proxy enrollment by volume, but Iran's 6.4 sessions/IP ratio — 3.4× the global average — could reflect fewer active nodes, heavier CGNAT, or both.

China leads by volume (558 IPs). India leads by fragmentation (315 IPs across 64 ISP organizations). Iran stands out with a higher sessions-per-IP ratio (6.4), which could reflect fewer heavily used proxy nodes, heavier carrier-grade NAT concentration, or a combination of both — CGNAT is prevalent on Iranian networks (particularly Irancell), meaning a single public IP may represent multiple devices behind a shared address.

India's mobile carriers Reliance Jio and Airtel together contribute 112+ unique IPs, consistent with compromised smartphones serving as proxy vectors. The top ISPs contributing attack traffic — Telmex (Mexico), HiNet (Taiwan), VNPT (Vietnam) — are legitimate consumer broadband providers serving millions of ordinary customers.

> **WHAT THIS INDICATES**
>
> There is no single country or network to block. Geographic blocking would cause unacceptable collateral damage against the same ISPs employees, customers, and partners use every day.

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The Supply Side

Two entirely separate ecosystems of compromised devices feed the residential proxy pipeline — and they share zero infrastructure.

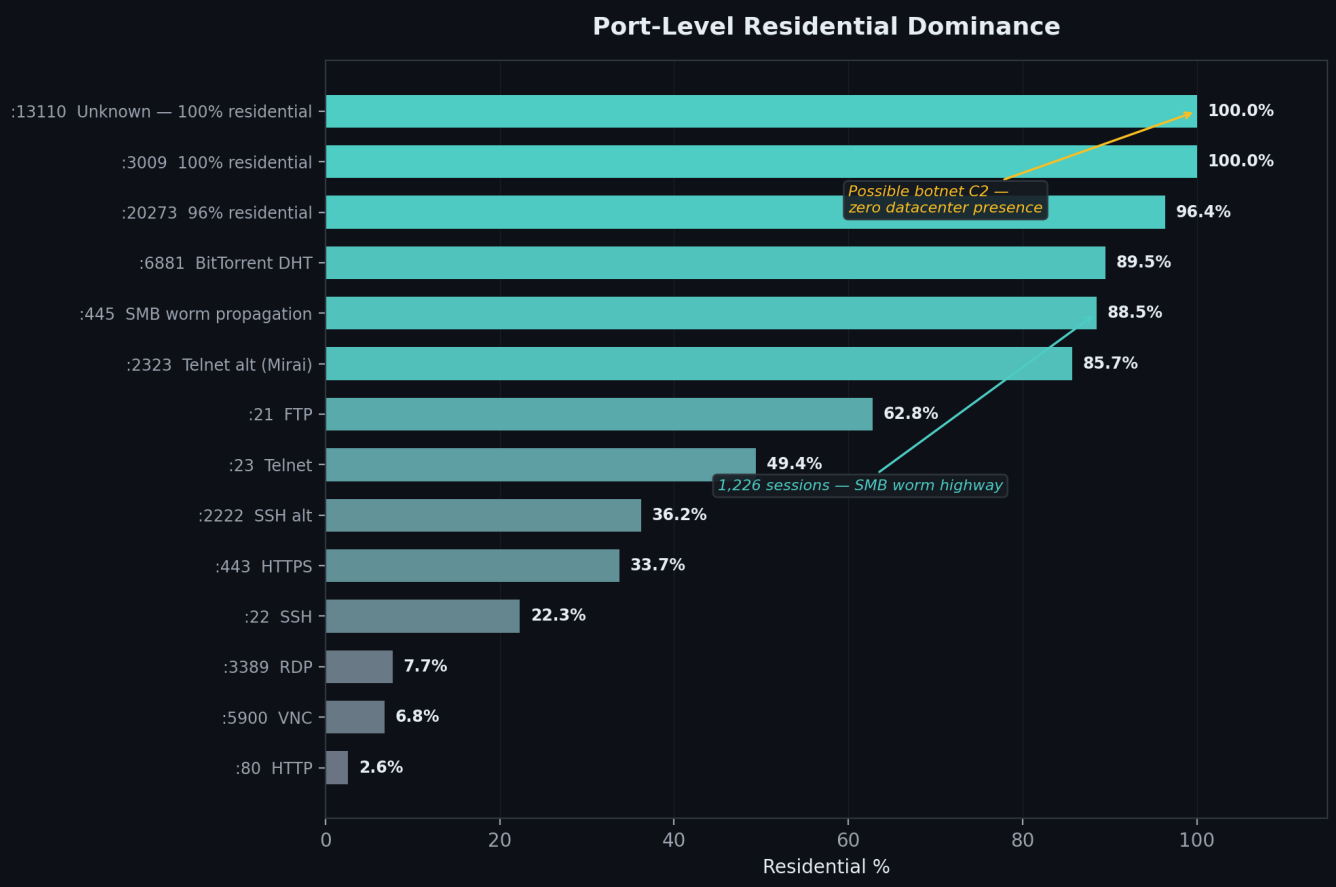

Figure 4: Port 13110 is 100% residential with zero datacenter presence across 115 IPs — a potential botnet command-and-control channel. Port 445 (SMB) at 88.5% residential confirms worm propagation via home networks.

SMB worm propagation runs 84% residential. Legacy worm infections — including Conficker variants (exploiting MS08-067, active since 2008) and EternalBlue-era variants (exploiting MS17-010, active since 2017) — self-propagate without any proxy operator involved. These machines scan the internet around the clock, for years, without their owners' knowledge. This is the single largest category of residential attack traffic — autonomous worms that have been propagating for over a decade. The devices they infect become the raw material of the residential proxy ecosystem.

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Meanwhile, consumer routers and cameras are recruited into botnets via default Telnet passwords. GreyNoise observes active recruitment campaigns — devices scanning for other vulnerable devices to enroll.

The zero-overlap finding is striking. Among 2,487 residential IPs, not a single one showed both SMB and Telnet activity. Under independence, we would expect approximately 22 IPs showing both SMB and Telnet activity. Observing zero in 2,487 IPs is statistically significant ($p < 10^{-9}$), confirming these are truly separate populations. Two completely independent supply chains feed two independent proxy networks. Defensive responses that address worm propagation do nothing for IoT recruitment, and vice versa.

> **WHAT THIS INDICATES**
> The residential proxy supply side is not one problem — it is at minimum two. Legacy worms infect PCs. Default credentials enable IoT device compromise. Both produce proxy-ready infrastructure, but through entirely different vulnerabilities and device populations.

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Recommendations

1. Block SMB/445 from ISP address space at the perimeter — there is no legitimate enterprise reason to accept inbound SMB from residential IPs, and 84% of residential attack traffic uses this protocol
2. Ensure IoT and OT devices cannot initiate outbound Telnet connections — compromised IoT devices use outbound Telnet to recruit other vulnerable devices into botnets; blocking outbound Telnet stops the propagation chain
3. Change all default credentials on internet-facing devices — default Telnet passwords are the primary vector for IoT botnet recruitment; credential changes eliminate the entry point entirely
4. Monitor for Telnet brute-force attempts targeting enterprise infrastructure — inbound Telnet credential stuffing indicates active botnet recruitment; early detection prevents enterprise devices from being enrolled as proxy nodes
5. Track JA4T fingerprint changes on ISP-classified traffic — commercial proxy SDKs update periodically, changing the TCP fingerprint of enrolled devices; tracking JA4T fingerprint changes over time on residential IPs reveals proxy enrollment that IP reputation misses, because the fingerprint survives IP rotation; flag PPPoE MSS 1452 on traffic targeting enterprise services as likely broadband-origin
6. Detect sequential single-touch rotation against authentication endpoints — when multiple ISP-classified IPs probe the same VPN login page or authentication service in sequence, each appearing only 1–2 times, this is proxy-routed reconnaissance; no individual IP triggers a threshold, but the aggregate pattern is a high-fidelity signal; detect the rotation pattern, not the individual IP

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Commercial Proxy Fleets

The hardest threat to detect. Commercial proxy networks enroll consumer devices as exit nodes through SDKs bundled into free apps — and the traffic looks identical to legitimate use.

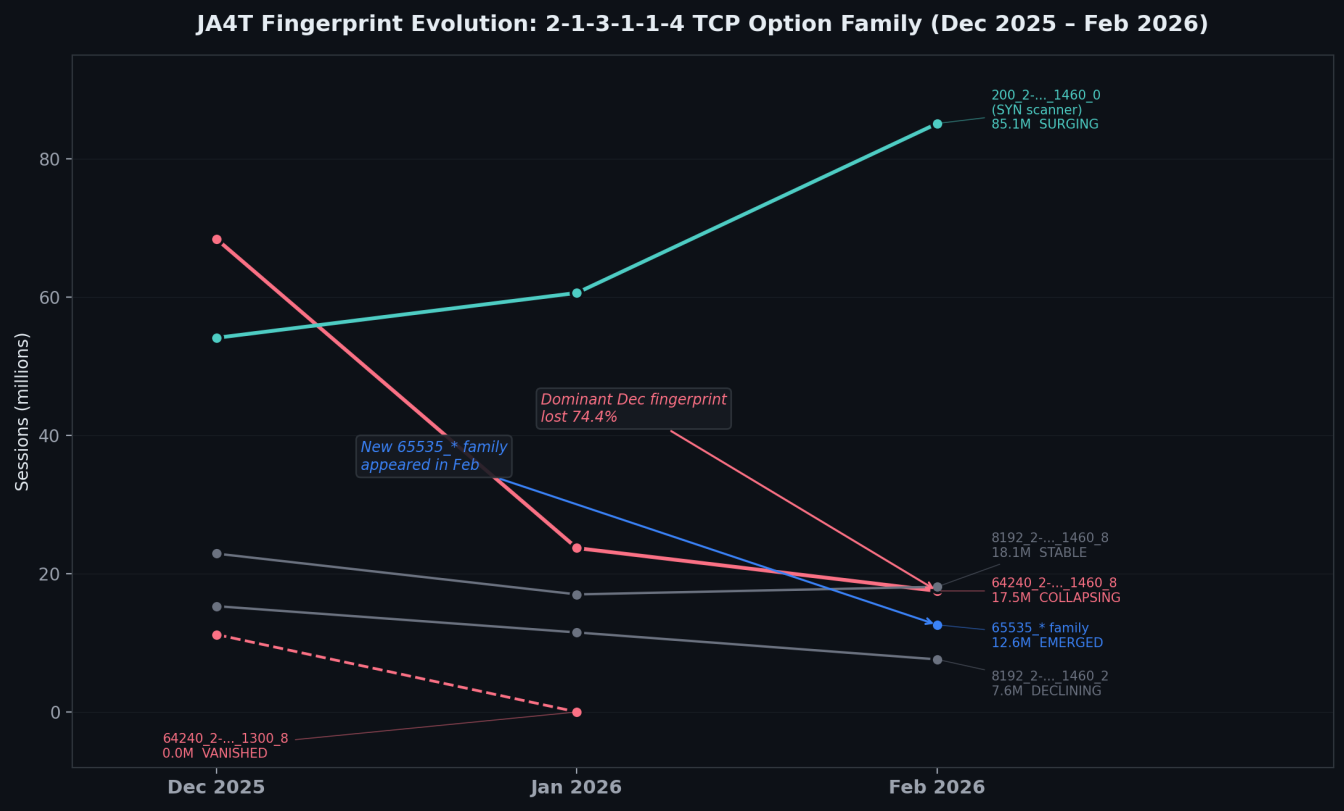

Figure 2: The standard Windows TCP option family is actively churning — the dominant December fingerprint collapsed 74% while new families emerged. Possible causes include SDK updates, Windows patches, ISP changes, or device turnover.

Networks like IPIDEA and 911 S5 work on a simple trade: the device owner gets a free VPN or ad-blocking app; the proxy operator sells their bandwidth to anyone willing to pay. These fleets show distinct technical signatures — device fingerprints spanning 36 countries that churn as the SDK is updated.

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According to Google Threat Intelligence Group, over 550 distinct threat groups used IPIDEA in a single week, including state-sponsored groups from China, DPRK, Iran, and Russia. Bright Data has pitched streaming operators to integrate proxy SDKs into Samsung Tizen and LG webOS smart TV apps — extending the proxy frontier into living room IoT.

> **WHAT THIS INDICATES**
> IP reputation alone cannot detect proxy-routed traffic. Device fingerprinting (JA4/JA4H) provides more durable detection because fingerprints survive IP rotation.

VPN Reconnaissance

Attackers are using residential proxies to map enterprise perimeters — and some of the IPs probing edge infrastructure run enterprise VPN software.

Among 33 residential IPs (1.3% of the residential sample) targeting enterprise VPN login pages — SonicWall, Citrix, Pulse Secure, and Ivanti gateways — each typically hitting a single vendor. This is proxy-routed reconnaissance at its sharpest: attackers deliberately using home internet connections to evade reputation-based blocking while inventorying corporate access points.

Four residential IPs attempted path traversal to `/bin/sh` — active exploitation routed through residential addresses. One IP ran Proxmox VE hypervisor credential stuffing (18 sessions) from a residential connection.

Enterprise VPN client signatures appeared on 48 residential IPs targeting the edge — 39 running FortiClient, 6 running GlobalProtect, and 3 running SonicWall VPN. This suggests compromised work-from-home endpoints, misclassified network appliances, or proxy malware co-resident with legitimate VPN software.

> *If your organization uses FortiClient VPN, audit whether any of your endpoints also exhibit scanning behavior. Work-from-home endpoints with proxy malware represent a direct path past your perimeter.*

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WHAT THIS INDICATES

Attackers route reconnaissance through residential IPs specifically to bypass reputation-based blocking. The traditional approach of blocking known bad IPs cannot work when the IPs look like employees' home connections.

When Networks Die

The residential proxy ecosystem is resilient by design. Take one network down and another absorbs the demand.

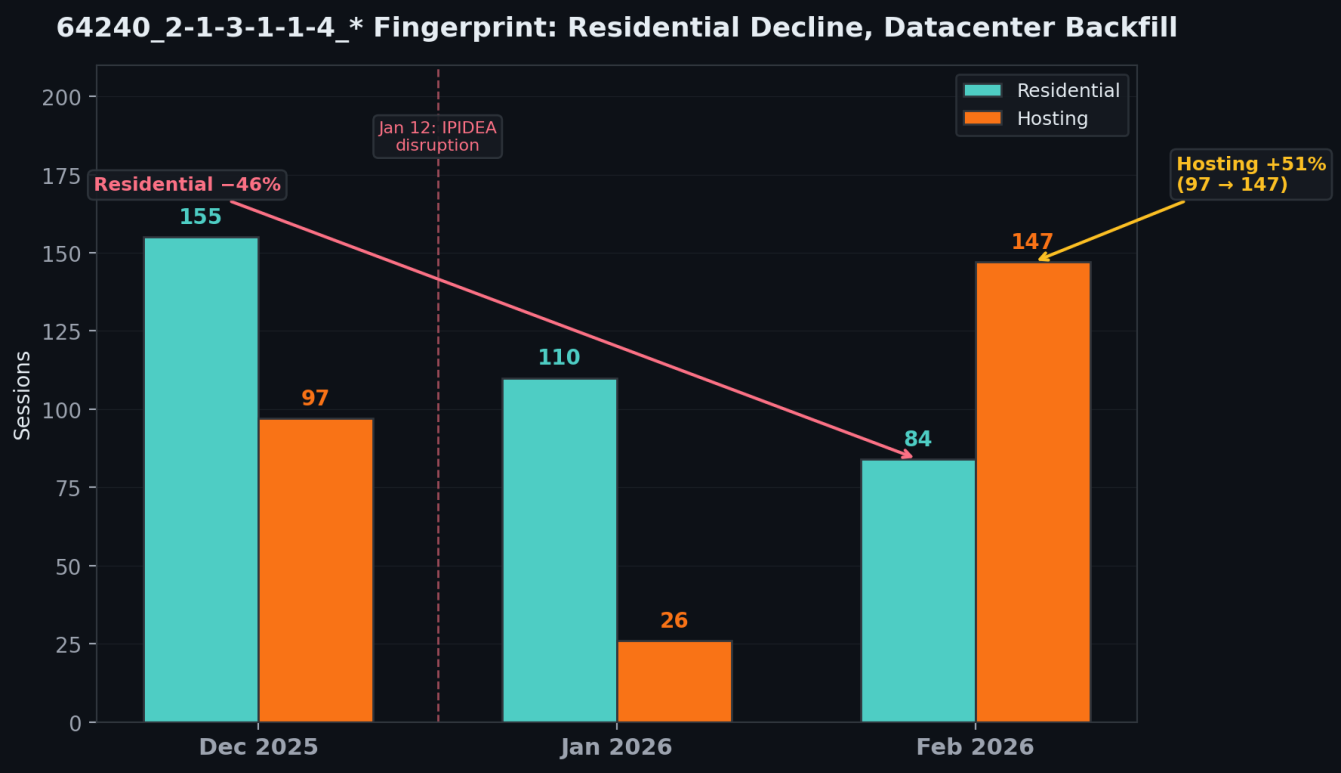

Figure 6: As residential sessions with the 64240_2-1-3-1-1-4_* fingerprint declined 46% from Dec to Feb, hosting sessions rebounded from 26 to 147 — temporally correlated with the Jan 12 IPIDEA disruption.

On January 29, 2026, Google Threat Intelligence Group disrupted IPIDEA — 9 to 11 million daily active proxies used by over 550 threat groups. The disruption reduced IPIDEA's pool by approximately 40%.

The timing is suggestive: residential sessions with fingerprints consistent with IPIDEA-linked infrastructure declined 46% from December to February, while hosting sessions rebounded in the

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same period — consistent with operators backfilling lost residential capacity with datacenter servers, though alternative explanations exist.

Date	Target	Scale	Outcome
Jan 2026	IPIDEA	9–11M daily proxies	Google-led disruption, ~40% reduction
May 2025	AnyProxy/5Socks	7,000+ proxies	4 indicted, $46M+ revenue
Nov 2024	NSOCKS/Ngioweb	35K daily bots	Exposed by Black Lotus Labs
May 2024	911 S5	19M IPs, 190 countries	FBI takedown, ~$99M revenue

Table 2: Law enforcement actions against residential proxy networks.

WHAT THIS INDICATES

Every major takedown produces the same result — temporary disruption, then regeneration. According to IPinfo research, 46% of proxy IPs appear across multiple providers simultaneously. 911 S5 was dismantled but briefly reappeared as CloudRouter. Disruption buys time; it does not solve the problem.

The Detection Gap

GreyNoise sees the supply side of the residential proxy ecosystem: botnet credential stuffing, worm propagation, and SSH brute-force that builds and maintains proxy infrastructure.

GreyNoise has limited detection for the demand side: traffic routed through residential proxies as exit nodes. There are no tags for known proxy networks (Ngioweb, IPIDEA, NSOCKS, Faceless), proxyware SDKs (PacketSDK, EarnSDK), or Operational Relay Box (ORB) network traffic patterns.

This gap is not unique to GreyNoise — it reflects a broader industry challenge. Residential proxy traffic is structurally difficult to detect because it blends with legitimate traffic at the network level.

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Specialized vendors including Spur, IPinfo's proxy detection service, and Synthient have developed detection capabilities specifically targeting residential proxy and proxyware traffic. These tools represent meaningful progress — particularly for identifying known proxy network fingerprints and proxyware SDK signatures. However, the structural challenge persists: residential IPs rotate faster than feed update cycles, newly enrolled IPs have no prior reputation, and novel SDKs evade signature-based detection until fingerprinted. The detection gap is narrower than MaxMind's 0% residential proxy detection rate suggests, but the rotation economy ensures it cannot be fully closed by any feed-based approach alone.

> **WHAT THIS INDICATES**
> The industry sees the worms that build proxy fleets but not the operations that use them. Behavioral analysis, device fingerprinting, and traffic pattern analysis are the most promising paths to closing this gap.

Forward Look

The residential proxy ecosystem will continue to adapt. Three trends warrant monitoring:

- Proxy fleet regeneration after disruption. The IPIDEA takedown demonstrated that operators backfill lost capacity within weeks. Future disruptions will likely produce similar patterns — temporary reduction followed by datacenter backfill and eventual residential re-enrollment.
- SDK expansion into new device categories. As noted above, Bright Data has begun targeting smart TV platforms — if successful, the residential proxy frontier extends from PCs and phones into living room IoT.
- State-sponsored adoption of residential proxy infrastructure. According to Google Threat Intelligence Group, over 550 distinct threat groups used IPIDEA in a single week. ORB networks used in the UNC3886 campaign targeting Singaporean telcos demonstrate how state actors leverage residential infrastructure for targeted operations.

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Methodology

Observation Period: November 29, 2025 — February 27, 2026 (90 days)

Data Source: GreyNoise Global Observation Grid — sensors deployed across 80+ countries observing unsolicited internet traffic

Dataset:

- 4,020,000,000 sessions (100% aggregation)
- 5,720,000 unique source IPs
- Excludes known benign scanners (Shodan, Censys, etc.)

Classification: IP type classification based on IPinfo data. "Residential" in this report refers to IPs classified as ISP-type, including home broadband, mobile data (Jio, Airtel, etc.), and small-business connections on ISP address space.

Sampling: Type distribution relies on a 30,000-session convenience sample (10K/month) because the IP type field is not aggregatable across the full dataset. Session share range: 15–30%. IP share (39%) is more stable and more reliable.

Validation: 50 residential IPs profiled via Censys — 42% genuinely residential by IP count, 62% including compromised CPE (MikroTik routers, Keenetic gateways). Behavioral evidence holds regardless of subpopulation composition.

Third-party sources cited:

- Google Threat Intelligence Group: IPIDEA disruption (January 2026)
- Mandiant M-Trends 2025: Scattered Spider, UNC3886 proxy usage
- DOJ: 911 S5 and AnyProxy/5Socks prosecutions
- Black Lotus Labs: NSOCKS/Ngioweb exposure (November 2024)
- IPinfo: IP type classification methodology

What GreyNoise observes: Our sensors detect scanning and probing attempts — traffic that reaches our honeypots also reaches real internet-facing infrastructure. We observe techniques and scale; we cannot confirm compromise of production systems.

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NAT and IP-to-device mapping: This analysis measures public IP addresses, not individual devices. Carrier-grade NAT (CGNAT), prevalent in India (Jio), Iran (Irancell), Brazil, and other countries with constrained IPv4 address space, means a single public IP may represent multiple devices. Sessions-per-IP ratios should be interpreted as upper bounds on per-device activity. Conversely, DHCP churn means a single device may appear under multiple public IPs over time. Both effects are inherent to IP-level measurement and do not affect behavioral findings (circadian patterns, protocol segmentation, rotation velocity).

Circadian analysis scope: The circadian analysis measures SMBv1 worm traffic (Windows-only). India's mobile-first device population (Jio, Airtel) would exhibit different idle patterns not captured by this analysis.

Honeypot observation bias: GreyNoise sensors are passive honeypots that capture scanning and reconnaissance traffic. Target-directed activities like web scraping, credential stuffing against specific services, and ad fraud are structurally invisible to this methodology. The residential proxy problem is larger than what any honeypot network can measure.

Key Indicators

The following structured indicators are derived from the analysis. For each category, we describe what defenders should look for and why it matters.

Ports & Protocols (Residential-Dominant)

- SMB/445 (84% residential — legacy worm propagation)
- Telnet/23 (IoT botnet recruitment)
- DCERPC/135 (worm lateral movement)

What to look for: Inbound SMB or DCERPC from ISP-classified address space is almost certainly worm propagation — not legitimate traffic. Inbound Telnet from residential IPs signals active botnet recruitment. Any of these protocols from non-hosting IPs warrants immediate investigation.

Fingerprints

- JA4T families associated with IPIDEA-linked proxy SDK [truncated]
- Linux TCP stacks on persistent SSH proxy endpoints (41 cross-month IPs)

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What to look for: Compare TCP fingerprints against the expected device type for an IP's ASN. A Linux TCP stack originating from a consumer ISP that normally serves Windows desktops suggests a compromised device or proxy endpoint. JA4T fingerprint changes on the same IP over time indicate SDK updates — a hallmark of commercial proxy enrollment.

Notable ASNs

- AS209334 (Modat B.V.) — scanner infrastructure
- Consumer ISPs: Telmex (Mexico), HiNet (Taiwan), VNPT (Vietnam), Reliance Jio (India), Airtel (India)

What to look for: High volumes of scanning or authentication attempts from major consumer ISPs. A single Reliance Jio IP hitting your VPN login page once is noise. Dozens of Jio IPs hitting it in sequence — each appearing only once — is proxy-routed reconnaissance.

VPN Client Signatures on Residential IPs

- FortiClient (39 IPs)
- GlobalProtect (6 IPs)
- SonicWall VPN (3 IPs)

What to look for: Enterprise VPN client user-agent strings originating from residential IPs targeting your infrastructure. This may indicate compromised work-from-home endpoints running proxy malware alongside legitimate VPN software. Cross-reference with your own VPN client inventory — if the IP is not an authorized endpoint, it is either stolen credentials or proxy-routed probing.

Behavioral Indicators

- Single-session IPs (57.4% of residential population)
- 1-2 session IPs (78% of residential population)
- Circadian traffic patterns aligned with local time zones (34% nocturnal dip, India)
- Zero protocol overlap between SMB and Telnet residential populations

What to look for: A pattern of single-touch IPs from residential address space hitting the same service in sequence — this is the rotation economy in action. No individual IP triggers a threshold, but the aggregate pattern is unmistakable. Alert on the pattern, not the IP: multiple residential IPs performing identical low-volume actions against the same target within a short window.

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About GreyNoise

GreyNoise tells you what's attacking everyone so you can focus on what's attacking you.

The Global Observation Grid processes ~500 million sessions daily across 5,000+ sensors in 80+ countries, identifying internet-wide attack activity in real time.

GREYNOISE

Website: greynoise.io

Check your IPs: check.labs.greynoise.io

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